

RAK19006 WisBlock Wireless Charge Module Datasheet

Overview

Description

The RAK19006 WisBlock Wireless Charge Module is designed to be a part of the battery charger. It is highly efficient, Qi-compliant, and has a single-chip wireless power receiver and charger. It integrates the receiver and linear charger and supports up to 5 W applications.

This board can be connected with the WisBlock Base board like RAK5005-O through the IO slot or two pogo pins. The chip supports wireless charge and direct battery charging.

Features

- WPC 1.2.4 compliant wireless power receiver
- JEITA compatible charge
- Ultra-low leakage current without transmitter
- Wireless and wired charge
- OVP, OCP, and OTP protection
- Supports up to 5 W applications
- Chipset: ConvenientPower CPS3008
- Module size: 15 X 25 mm

Specifications

Overview

Mounting

The RAK19006 module can be mounted on the IO slot of the WisBlock Base board. **Figure 1** shows the mounting mechanism of the RAK19006 on a WisBlock Base module, such as the RAK5005-O.

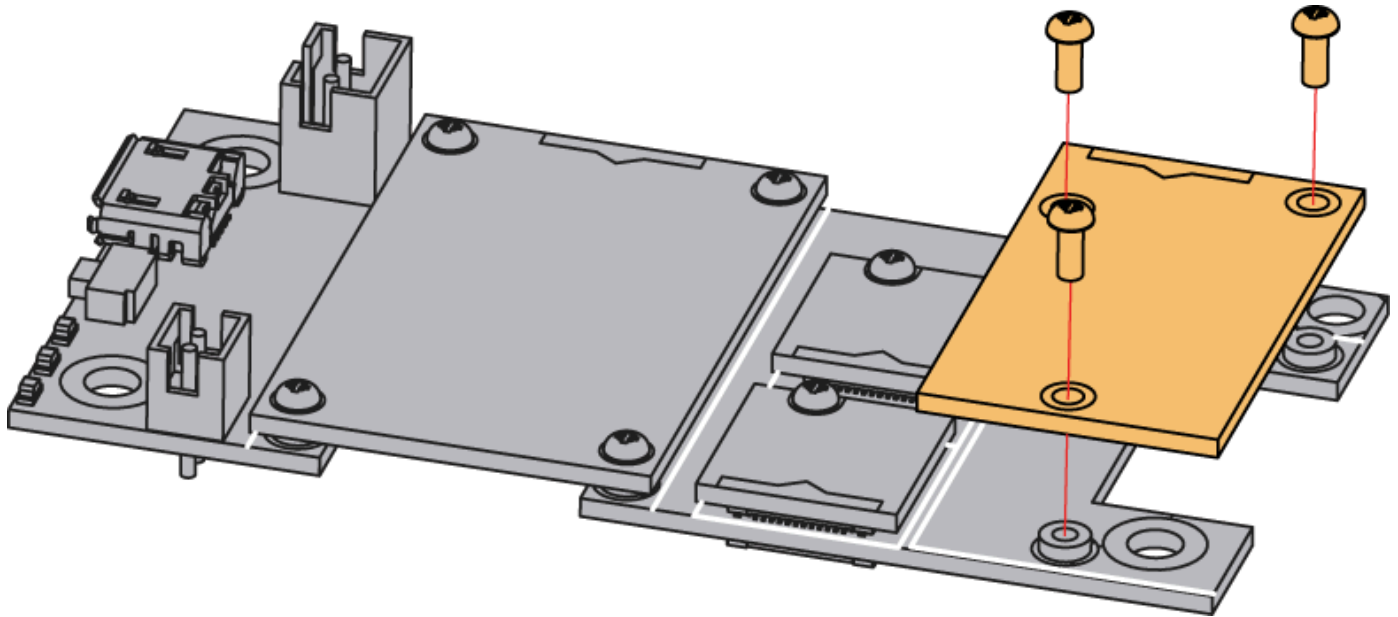


Figure 1: RAK19006 mounting mechanism on a WisBlock Base module

Hardware

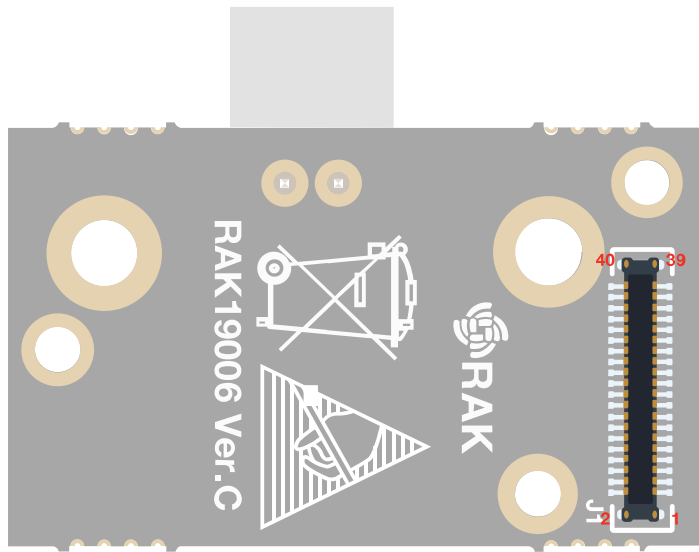
The hardware specification discusses the pinouts, and its corresponding functions and diagrams of the module. It also covers the electrical and mechanical characteristics that include the tabular data of the functionalities and standard values of the RAK19006 WisBlock Module.

Chipset

Vendor	Part number
ConvenientPower	CPS3008

Pin Definition

The RAK19006 module has a 40-pin WisConnector that is compatible to the WisBlock Base IO Slot. The pin order of the connector and the pinout definition is shown in **Figure 2**.



VBAT	2	1	VBAT
GND	4	3	GND
NC	6	5	NC
NC	8	7	NC
NC	10	9	VBUS
NC	12	11	NC
NC	14	13	NC
NC	16	15	NC
NC	18	17	NC
NC	20	19	NC
NC	22	21	NC
NC	24	23	NC
NC	26	25	NC
NC	28	27	NC
NC	30	29	NC
NC	32	31	NC
NC	34	33	NC
NC	36	35	NC
NC	38	37	NC
GND	40	39	GND

Figure 2: RAK19006 Pinout Schematic

NOTE

- Only **GND** and **nEN** are connected to this module.
- nEN** is the wireless charge chip enable pin, active low. A resistor (R4) is connected to nEN to GND, so by default, this chip is enable. If you want to disable it, pull the nEN to high.

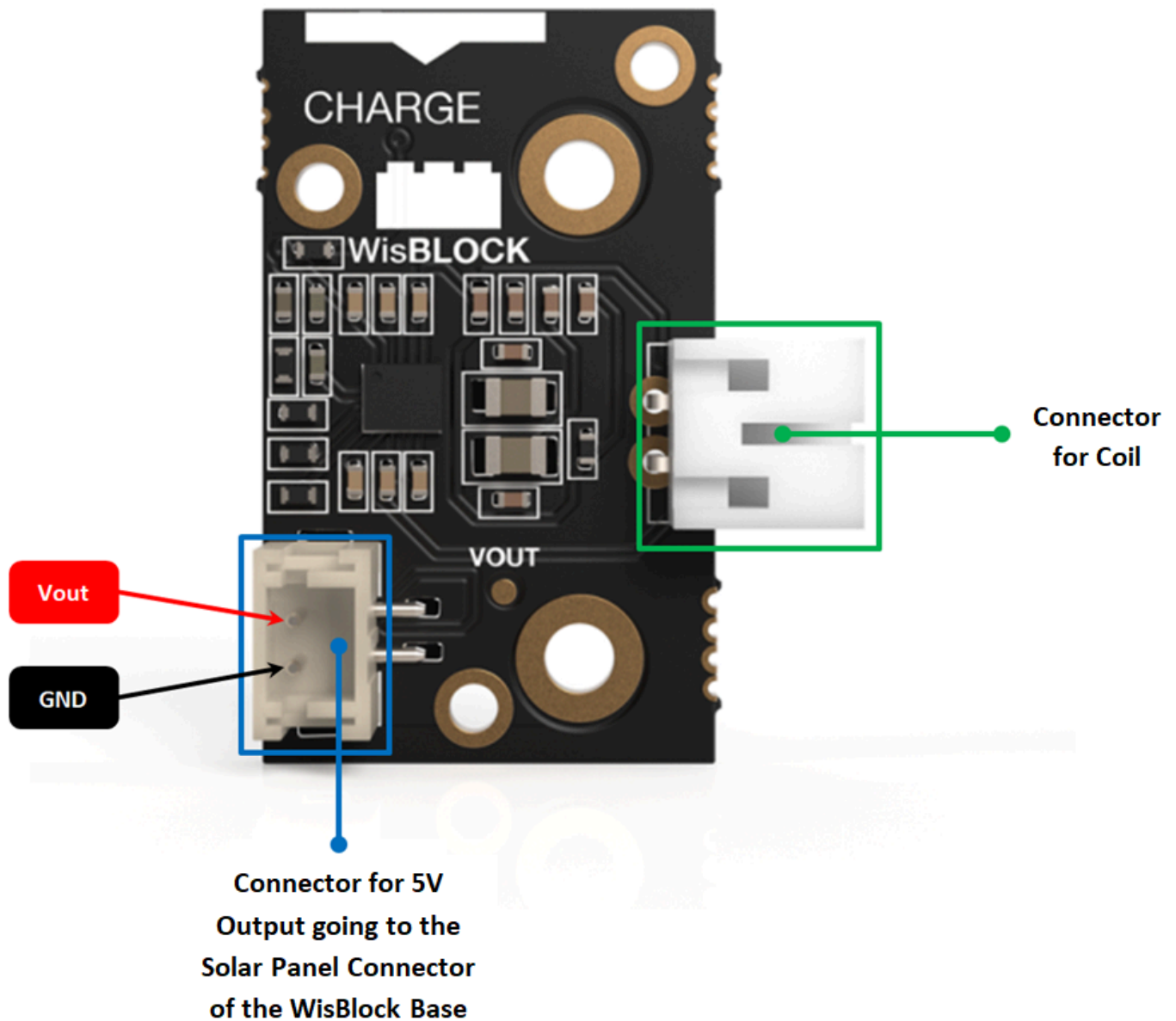


Figure 3: RAK19006 Pinout and Connector assignments

Electrical Characteristics

This sections shows the maximum and minimum ratings of the RAK19006 module and its recommended operating conditions. Refer to the table presented below.

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VRECT	Power voltage range	3.4	-	16	V

Symbol	Description	Min.	Nom.	Max.	Unit
Vout	Output voltage range	-	5	5.5	V
Iout	Continuous output current	-	0.6	1.1	A
TA	Ambient operation temperature	-40	-	85	°C

Mechanical Characteristic

Board Dimensions

The mechanical dimensions of the RAK19006 module is shown in **Figure 3** below.

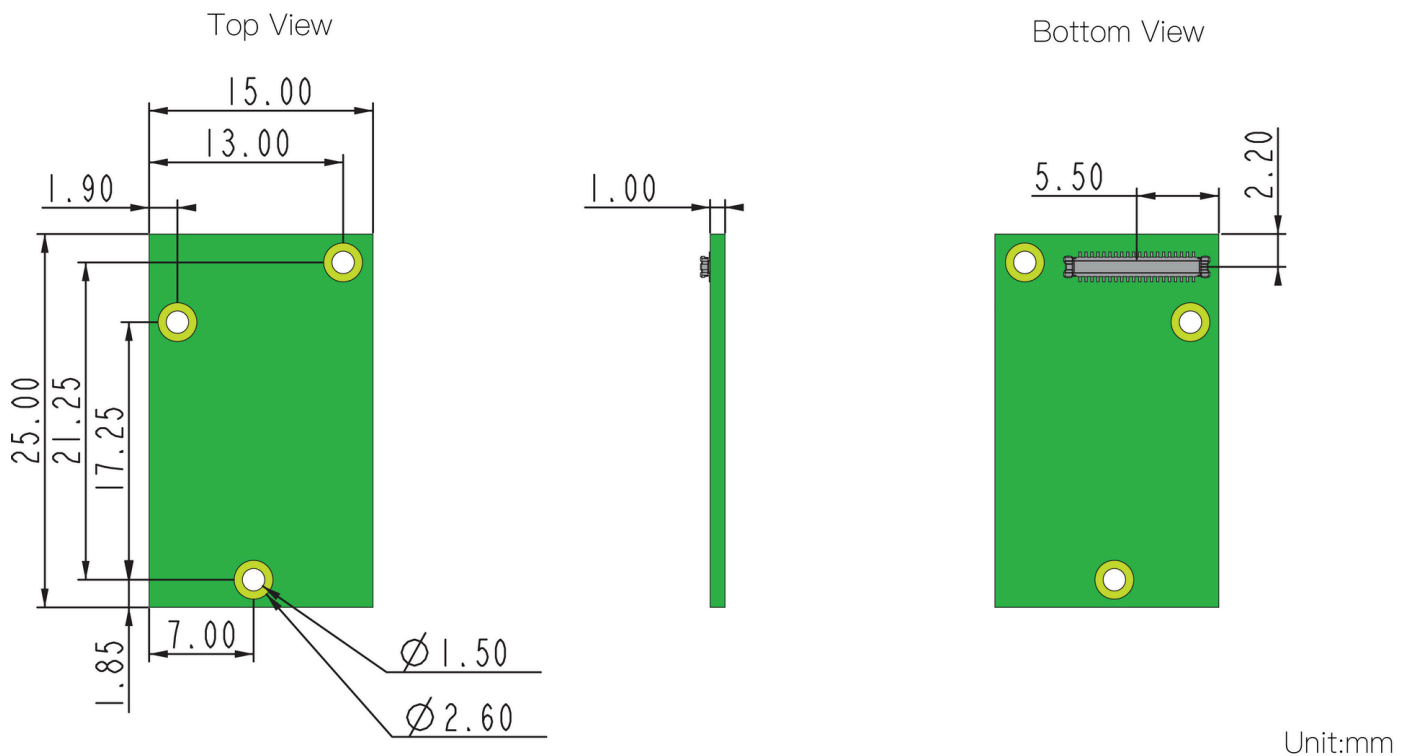


Figure 4: RAK19006 Mechanical Dimensions

WisConnector PCB Layout

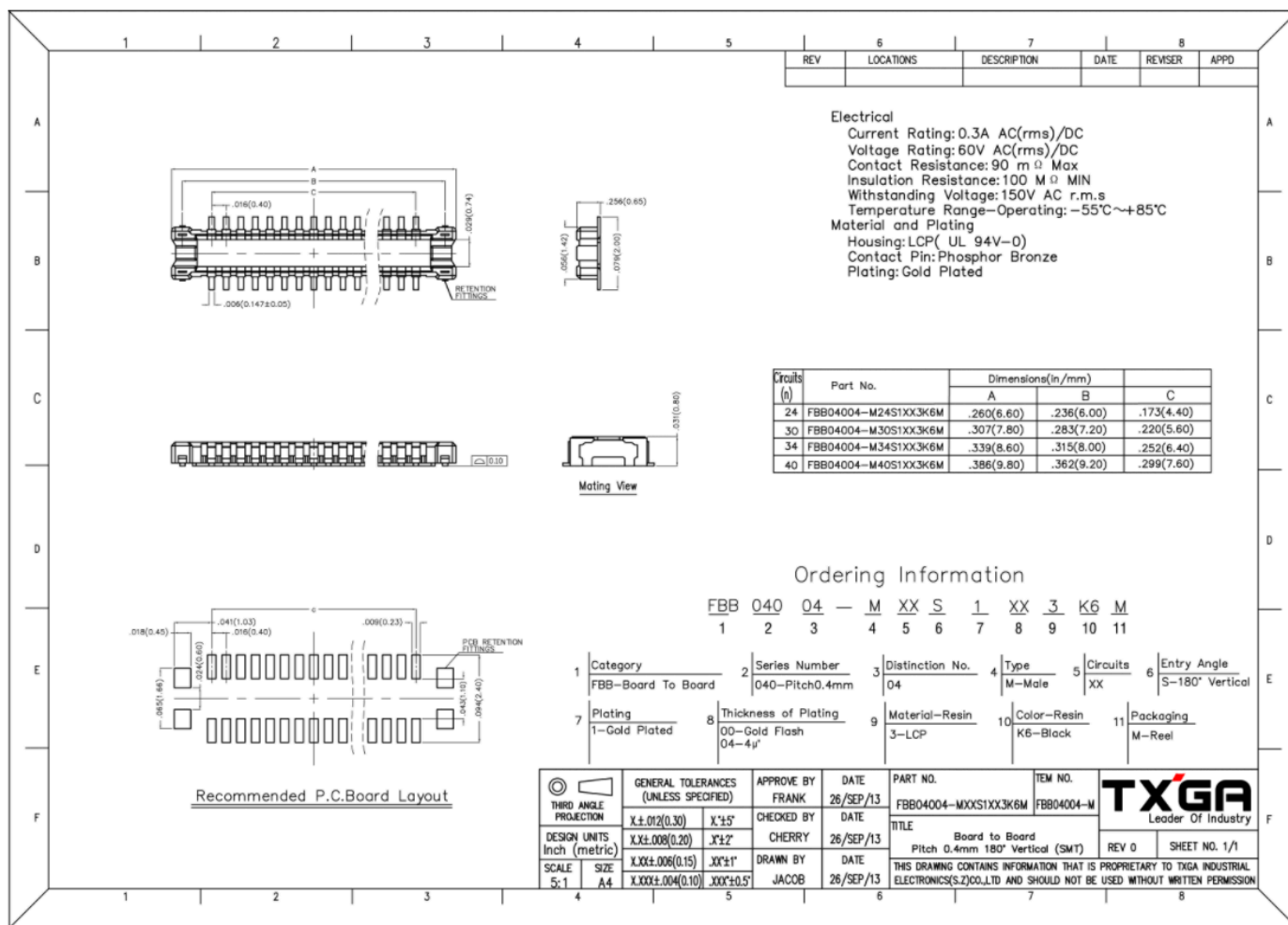


Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 5 shows the schematic of the RAK19006 module.

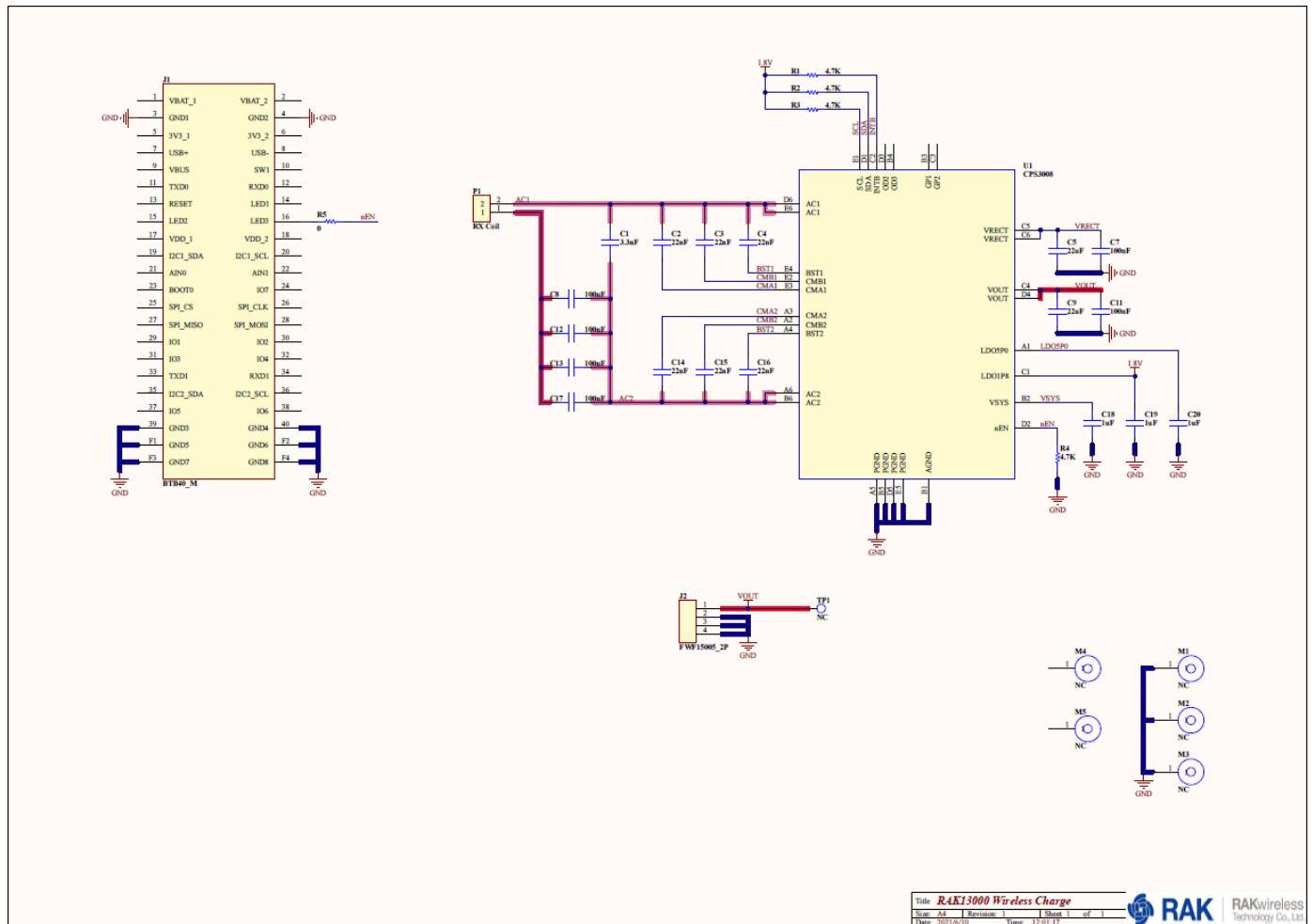


Figure 6: RAK19006 WisBlock Module Schematics

Charge chip

Figure 6 shows the schematic of RAK19006 wireless charge chip, J3 is a wireless charge coil connector, and VOUT is a wireless charge chip output voltage, with a normal voltage of 5 V.

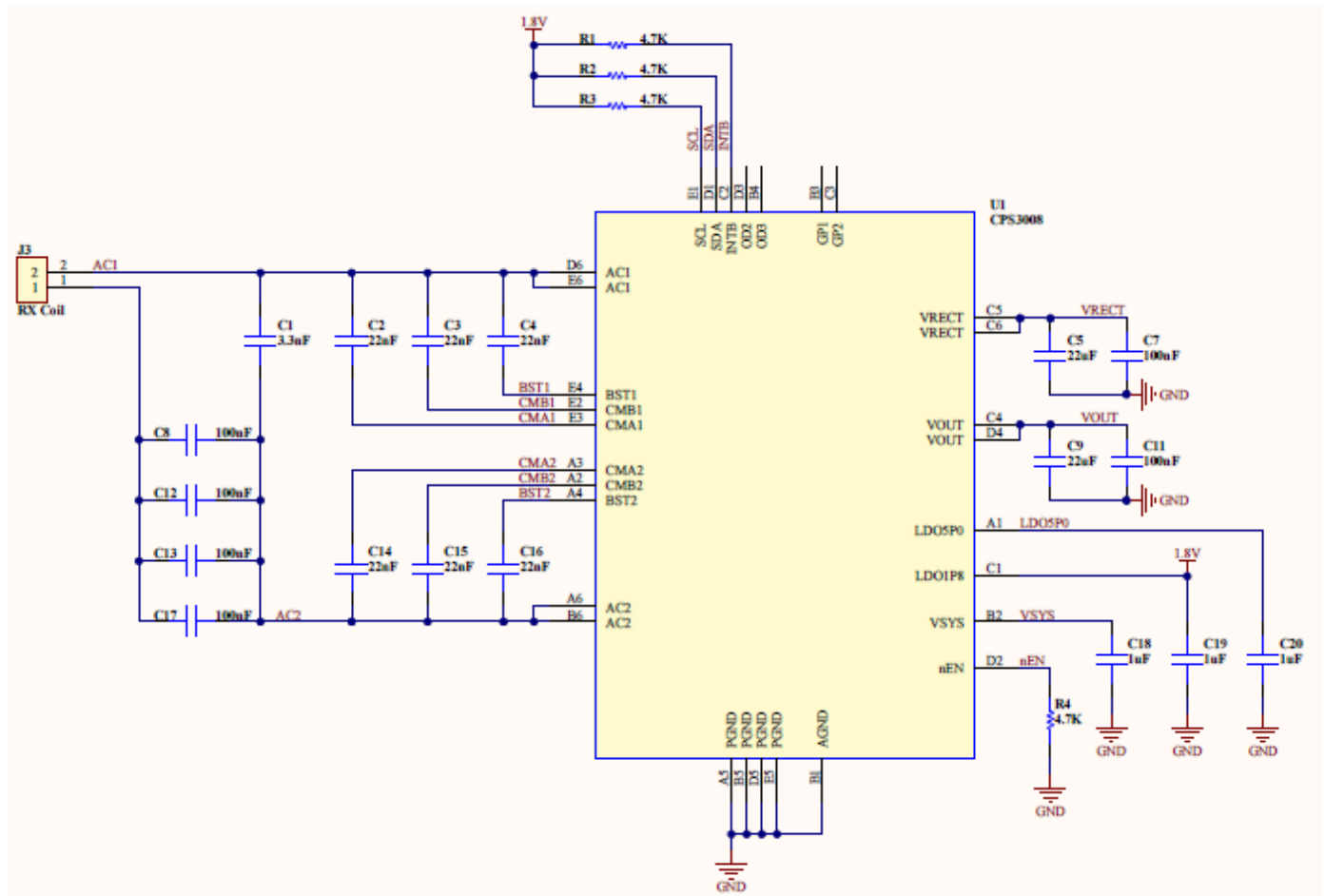


Figure 7: RAK19006 WisBlock Wireless Charge Module Schematic

Output Connector

Figure 7 shows the 2-pin cable wireless charge output connector connected to the WisBase solar input connector.

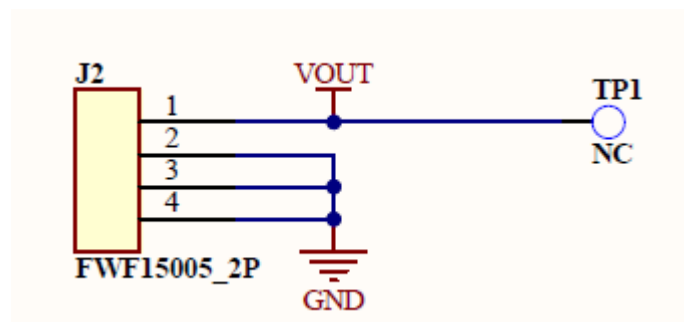


Figure 8: RAK19006 WisBlock Wireless Charge Output Connector

NOTE

- J2 is a 2-pin connector, Pin 3 and Pin 4 are the mounting pads.

